ASSIGNMENT NO 8 SY

Title

Extraction, Cleaning, Storage, and Reloading of Historical Rainfall Data from Data.gov.in

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Aim

To scrape tabular historical rainfall data from the *Data.gov.in – Rainfall India* resource, clean missing values, store the cleaned dataset into a CSV file, and reload the data for subsequent analysis.

Prerequisites

Before performing the lab work, students should have:

1. **Basic Python knowledge**
2. **Understanding of HTML & web scraping principles**
3. Python libraries installed:
    - requests
    - BeautifulSoup
    - pandas

Install required libraries with:

pip install requests beautifulsoup4 pandas

Objective

By the end of this lab session, you should be able to:

✔ Scrape structured rainfall data from an online source
✔ Load scraped data into a Pandas DataFrame
✔ Identify and handle missing values
✔ Save cleaned data to CSV format
✔ Reload CSV data for analysis

Theory

Rainfall data is critical for agricultural planning, hydrological studies and climate research. **Web scraping** allows us to extract data from websites that do not provide it in a direct database or API format.

In this lab, you will:

1. Use requests to fetch HTML source of the given dataset page.
2. Parse HTML with BeautifulSoup to extract *tabular data*.
3. Load data into a pandas.DataFrame.
4. Clean missing and invalid values using DataFrame methods.
5. Store the final DataFrame to a **CSV file**.
6. Reload the CSV file to verify storage and prepare for analysis.

Result

You have successfully:

✔ Scraped rainfall data
✔ Cleaned and formatted the dataset
✔ Stored the data into a CSV file
✔ Reloaded the CSV for analysis

Conclusion

In this lab exercise, we learned how to extract tabular data from an online source using web scraping, clean the dataset to handle missing and inconsistent values, store the cleaned data into a CSV file, and reload it for further usage.

This process is widely used in *real-world data science* for gathering datasets from public portals that do not provide direct APIs or downloads.